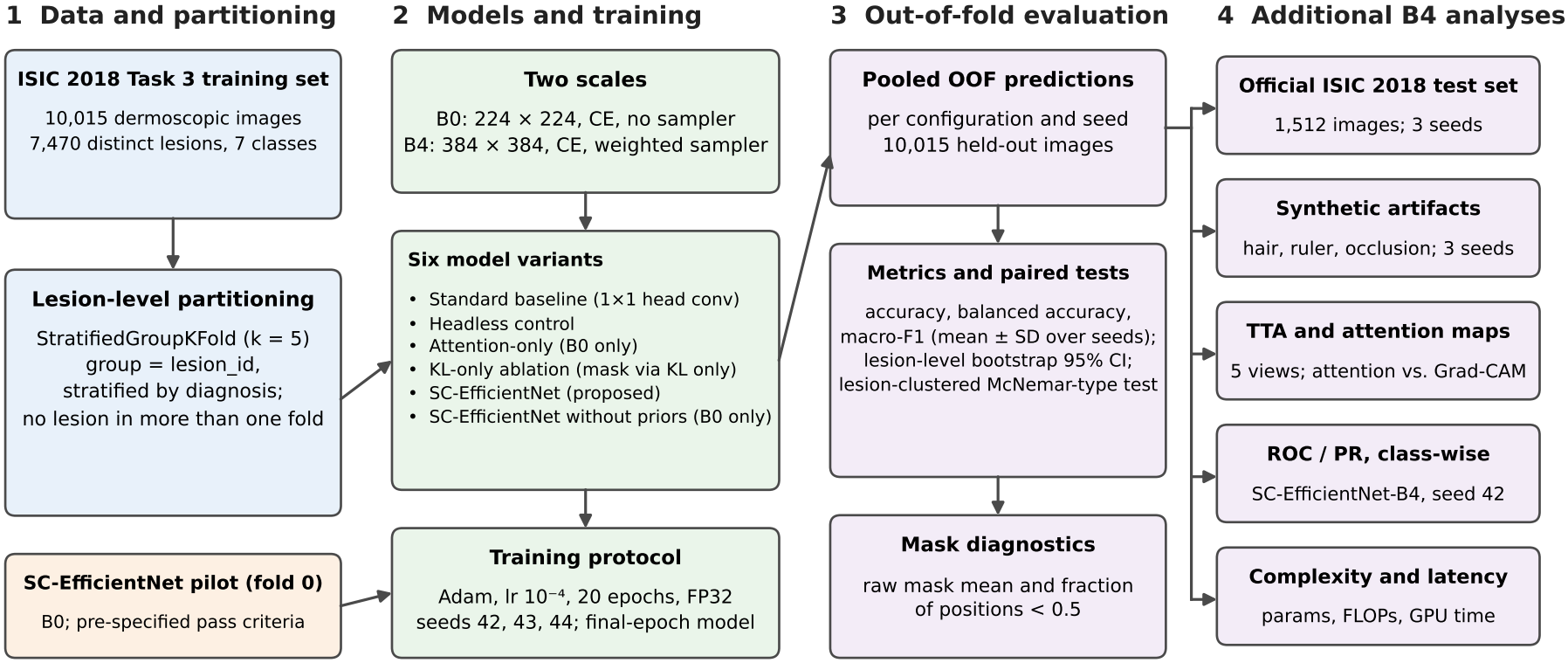
1 Data and partitioning
ISIC 2018 Task 3 training set
10,015 dermoscopic images
7,470 distinct lesions, 7 classes
Lesion-level partitioning
StratifiedGroupKFold (k = 5)
group = lesion_id,
stratified by diagnosis;
no lesion in more than one fold
SC-EfficientNet pilot (fold 0)
B0; pre-specified pass criteria
2 Models and training
Two scales
B0: 224 × 224, CE, no sampler
B4: 384 × 384, CE, weighted sampler
Six model variants
Standard baseline (1×1 head conv)
Headless control
Attention-only (B0 only)
KL-only ablation (mask via KL only)
SC-EfficientNet (proposed)
SC-EfficientNet without priors (B0 only)
Training protocol
Adam, lr 10⁻⁴, 20 epochs, FP32
seeds 42, 43, 44; final-epoch model
3 Out-of-fold evaluation
Pooled OOF predictions
per configuration and seed
10,015 held-out images
Metrics and paired tests
accuracy, balanced accuracy,
macro-F1 (mean ± SD over seeds);
lesion-level bootstrap 95% CI;
lesion-clustered McNemar-type test
Mask diagnostics
raw mask mean and fraction
of positions < 0.5
4 Additional B4 analyses
Official ISIC 2018 test set
1,512 images; 3 seeds
Synthetic artifacts
hair, ruler, occlusion; 3 seeds
TTA and attention maps
5 views; attention vs. Grad-CAM
ROC / PR, class-wise
SC-EfficientNet-B4, seed 42
Complexity and latency
params, FLOPs, GPU time